

A PROJECT TITLE
ON
**Breast Cancer Classification Using Decision Trees and Random Forest with ROC-
AUC/PR-AUC Evaluation**

project report submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING
BACHELOR OF TECHNOLOGY
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by

Name : K.V.Koushal Roll Number: 2520030009

Name : Bhavitha Reddy Roll Number: 2520030120

Under the Esteemed guidance of

Shaik. Asif

Assistant Professor, CSE



Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)

Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.

Phone No: 7815926816, www.klh.edu.in



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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Name of Title: Breast Cancer Classification Using Decision Trees and Random Forest with ROC-AUC/PR-AUC Evaluation

1. Abstract:

Breast cancer is one of the most common and life-threatening diseases affecting women worldwide, making early and accurate diagnosis essential for effective treatment. This project develops a machine learning-based classification system to distinguish between benign and malignant breast tumors using the Wisconsin Diagnostic Breast Cancer (WDBC) dataset, which contains diagnostic features extracted from digitized fine needle aspirate (FNA) images, including radius, texture, perimeter, area, and smoothness of cell nuclei. The workflow involves data preprocessing through handling missing values, feature scaling, and splitting the dataset into training and testing sets. Two tree-based machine learning models, Decision Tree and Random Forest, were implemented and compared for binary classification. The Decision Tree served as the baseline model, while the Random Forest combined multiple decision trees to reduce overfitting and improve generalization. Model performance was evaluated using Accuracy, Receiver Operating Characteristic Area Under Curve (ROC-AUC), and Precision-Recall Area Under Curve (PR-AUC), with the latter two providing a more reliable assessment for medical datasets that may exhibit class imbalance. The comparative analysis showed that the Random Forest achieved superior predictive performance, demonstrating higher robustness and better classification capability than the single Decision Tree. Additionally, feature importance analysis was performed to identify the most influential diagnostic attributes, improving the interpretability of the model. Overall, the project highlights the effectiveness of ensemble-based tree methods and clinically relevant evaluation metrics in developing a reliable and interpretable decision-support system for early breast cancer diagnosis.

2. INTRODUCTION:

Breast cancer is one of the leading causes of cancer-related deaths among women worldwide, making early and accurate diagnosis essential for improving treatment outcomes and patient survival. Machine learning has emerged as a powerful tool for assisting healthcare professionals by providing faster, more consistent, and data-driven diagnostic support. Traditional diagnostic methods often rely on manual examination of biopsy samples, which can be time-consuming and subject to human error or variation among medical practitioners. This project investigates the use of two widely adopted supervised machine learning algorithms, Decision Tree and Random Forest, for the classification of breast tumors into benign and malignant categories using the Wisconsin Diagnostic Breast Cancer (WDBC) dataset. The Decision Tree algorithm offers a simple and highly interpretable model by making classifications through a sequence of feature-based decisions, making it suitable for medical applications where transparency is important. However, single decision trees are prone to overfitting and may not generalize well to unseen data. To overcome this limitation, the Random Forest algorithm constructs an ensemble of multiple decision trees using random subsets of data and features, thereby improving prediction accuracy, reducing variance, and enhancing model robustness. The performance of both models is evaluated using Accuracy, Receiver Operating Characteristic Area Under Curve (ROC-AUC), and Precision-Recall Area Under Curve (PR-AUC). While accuracy provides an overall measure of correctness, ROC-AUC evaluates the model's ability to distinguish between benign and malignant tumors across different classification thresholds, and PR-AUC offers a more informative assessment of the model's

effectiveness in identifying malignant cases, particularly in datasets with class imbalance. Additionally, feature importance analysis is performed to identify the diagnostic attributes that contribute most significantly to classification decisions, improving model interpretability. Overall, this project aims to develop and compare reliable machine learning models that can serve as effective decision-support tools for the early detection and diagnosis of breast cancer.

3. SOFTWARE REQUIREMENTS:

Programming Language: Python 3.x

IDE/Environment: Jupyter Notebook / Google Colab / VS Code

Libraries: pandas, numpy — data handling and preprocessing

scikit-learn — Decision Tree, Random Forest, train-test split, metrics (roc_auc_score, average_precision_score, etc.)

matplotlib, seaborn — data visualization and plotting ROC/PR curves

Operating System: Windows/Linux/macOS (any OS supporting Python)

Hardware Requirements:

Processor: Intel i3 or above (i5 recommended)

RAM: Minimum 4 GB (8 GB recommended for smoother processing)

Storage: Minimum 500 MB free disk space

Display: Standard monitor (1366x768 or higher resolution)



Signature of the Guide



Course Instructor